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E E 476 A
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CAD 5 FINAL PROJECT: REGISTER FILE

System Overview and Architecture. Our design implements a 16-bit, 13-entry Register File (Figure 1 and Figure 10) using a top-down approach. The high-level architecture consists of three primary components: a Decoder for write operations, Multiplexers (Muxes) for read operations, and an array of Bit Slices that store the data. The write process utilizes a 4-to-16 decoder to map a 4-bit write address to 16 enable lines, while the read process uses multiplexers to select specific data bits based on read addresses. As shown in Figure 1 and Figure 10, the system is designed to handle dual read capabilities, requiring two separate read addresses (Read Address 0 and Read Address 1) and corresponding data outputs.

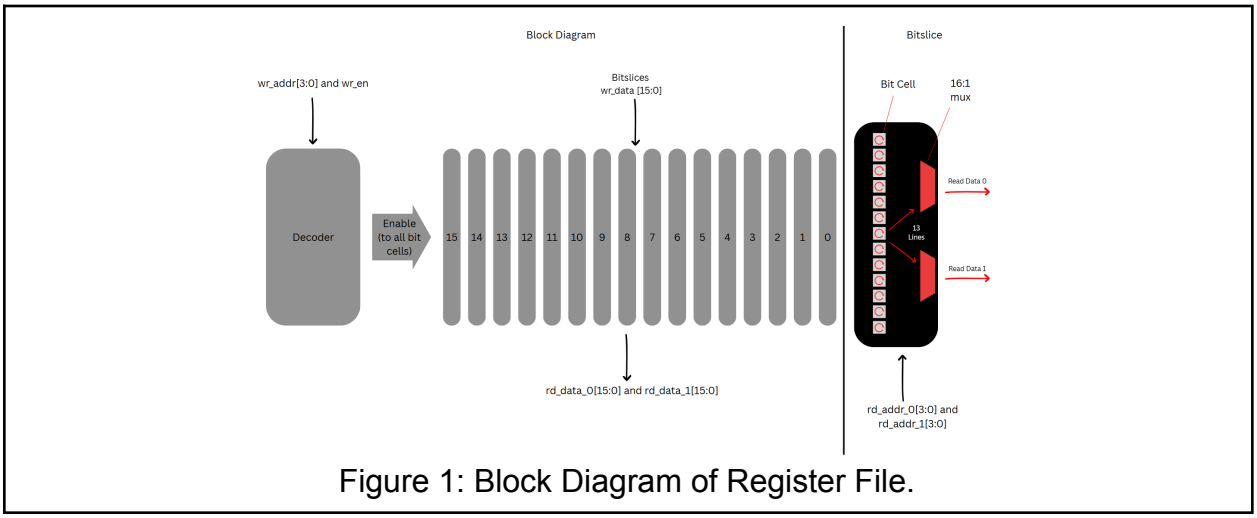


Figure 1: Block Diagram of Register File.

Circuit Design and Implementation.

Bit Cell Design. The fundamental unit of storage, the bit cell, is built around a Master-Slave Flip-Flop configuration, leveraging our team's familiarity with the DFAR schematic (Figure 2). To manage data storage and writing, we implemented a control mechanism using two transmission gates. The first transmission gate connects the write input to the flip-flop and is controlled by an enable signal; when the row is enabled, this gate opens to allow new data to be written. The second transmission gate creates a feedback loop between the input and output, remaining open when the write enable is off to maintain the stored bit state.

Decoder Design. We utilized a Decoder (Figure 5) to manage the addressing for the write process, mapping 4 input bits to 16 row enable lines. The write address and an enable signal serve as inputs to the Decoder; when enabled, the Decoder activates 1 of the 16 outputs based on the address provided. Although the component supports 16 lines, we utilize 13 of these signals to connect to the 13 rows of the Register File. This

ensures that we write to only one row at a time; when a specific row is enabled, all 16 Bit Cells in that row become accessible for writing. Figures 2 and 3 show sub-schematics utilized in the Decoder design.

Mux Design. The Mux (Figure 8) is used to encode the 16 row signals back to 4 and determine the read output. The read address inputs function as the select bits for the Mux, choosing the specific Bit Cell or row to read from within a Bit Slice. Figures 6 and 7 show sub-schematics of the Mux. Since the system requires dual read capabilities, each column must support two independent read paths

Bit Slice Design. A single Bit Slice (Figure 9) represents one column of data and contains two Muxes to accommodate the two separate read addresses. A column of 13 Bit Cells feeding into these two Muxes makes up a complete Bit Slice. All Read 0 and Read 1 Muxes share the same inputs from the Bit Cells, and their outputs provide the 16 bits currently being read (Figure 9). Any bits not being read are preserved and cycled by their respective Bit Cells. Because only one row is enabled at a time for writing, we tied every input of each Bit Cell together in a Bit Slice and designated it as a specific bit number (Figure 1) (Figure 9). The final Register File is composed of 16 of these Bit Slices to store the required 16 bits for each of the 13 entries.

Trade Offs: Power, Performance, Area.

Gate Sizing. Our gates were designed with a pitch of 1.4μm, incorporating a Wn/Wp ratio of 450/300 (1.5). While standard optimal Wn/Wp ratios typically range from 2 to 3, we determined that a ratio of 1.5 remained fully functional while allowing us to minimize the size of our active regions. Furthermore, by standardizing the pitch and width of individual logic gates (such as NAND2 and NOR2), we were able to tile these gates into larger hierarchical components, like the Bit Slice, with significantly greater ease.

Power and Performance. To optimize the power usage of the Register File circuit, we prioritized symmetry throughout the layout. This approach minimizes device mismatch and ensures that parasitic capacitances generated during the layout process are balanced. Additionally, we focused on optimizing wire routing to ensure minimum wire usage; reducing the length of implemented wires directly reduces the parasitic capacitance, thereby improving overall performance.

Area and Final Fit. The final dimensions of the Register File were measured at 171.11μm by 45.97μm, which fits well within the specifications outlined in the project requirements. We acknowledged that the layout could have been further optimized for length/width symmetry—for example, by implementing the Decoder physically above the Bit Slices. However, we chose the current implementation to prioritize layout simplicity, significantly reducing the risk of LVS errors and accidental shorting.

Hierarchy and Testing. To balance workload and complexity, we prioritized efficiency by selecting a Master-Slave Flip-Flop configuration. This allowed us to leverage our familiarity with the DFAR schematic and repurpose verified components (i.e. inverters

and transmission gates) to reduce debugging time. To manage complexity, we employed a hierarchical workflow—for example, constructing the Mux from three levels of cells—and validated each cell individually before layout. This modular approach ensured that the final integration required wiring only 17 pre-verified instances, giving us high confidence in the system's functionality.

All circuits underwent rigorous testing prior to integration. We utilized waveform testbenches to verify operations at every stage, from individual gates to the full Register File. The Appendix includes these testbenches, and Table 1 correlates the specific schematics with their corresponding testbench figures.

Table 1: Schematic Names and Corresponding Circuits and Testbenches.

Name	**Schematic**	**Testbench**
Bit Cell	Figure 2	Figure 11
DECODER16	Figure 5	Figure 12
MUX16	Figure 8	Figure 13
Bit Slice	Figure 9	Figure 14

Measurements. Measurements were conducted as specified in the project instructions, with energy calculations averaged over a minimum of five clock cycles. The pre-layout (schematic) Register File yielded an average read energy of 4.411 pJ and a write energy of 4.409 pJ. For the post-layout (extracted) simulation, these values increased slightly to 4.620 pJ for read operations and 4.615 pJ for write operations.

Frequency was determined by performing sweeps and analyzing the waveforms to identify the point of failure. The pre-layout schematic achieved a maximum operating frequency of 1.5 GHz, while the post-layout design operated successfully at 1.0 GHz, both of which meet the specifications set forth by the project requirements. The observed increase in energy consumption and decrease in maximum frequency in the post-layout simulation are expected results of the C+CC extractions. The inclusion of parasitic capacitances and resistances from the physical layout increases the load on the circuit, requiring more energy to drive signals and introducing delays that limit the maximum switching speed.

Contributions. From December 2nd through December 9th, the team dedicated over 38 hours to the project. Aiden Sallows led the physical design, completing layouts for the DFAR, Bit Cell, Bit Slice, Decoder, Mux, and RF, while also contributing to the Mux schematic, RF testing, and layout report section. Mukta Mishra was responsible for the schematics and testing of the Bit Cell, Bit Slice, Decoder, and RF, in addition to authoring the majority of the final report. Amita Prabhala managed the DFAR and Mux schematics, testing for the Mux, Bit Slice, and RF, wrote the testing section of the report, and provided standard cell layouts from previous assignments

Appendix.

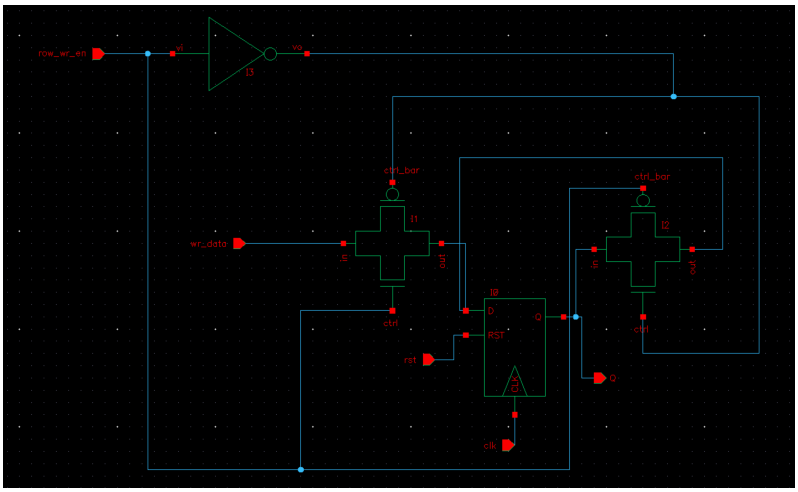


Figure 2: Bit Cell Schematic.

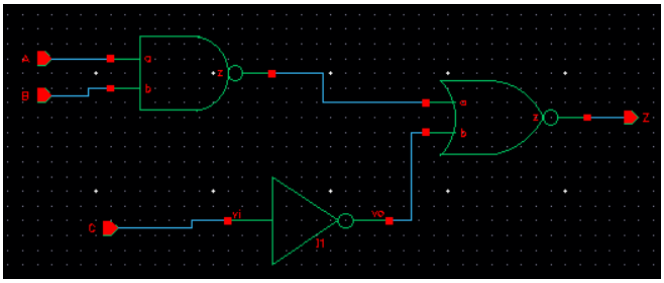


Figure 3: AND3 Schematic.

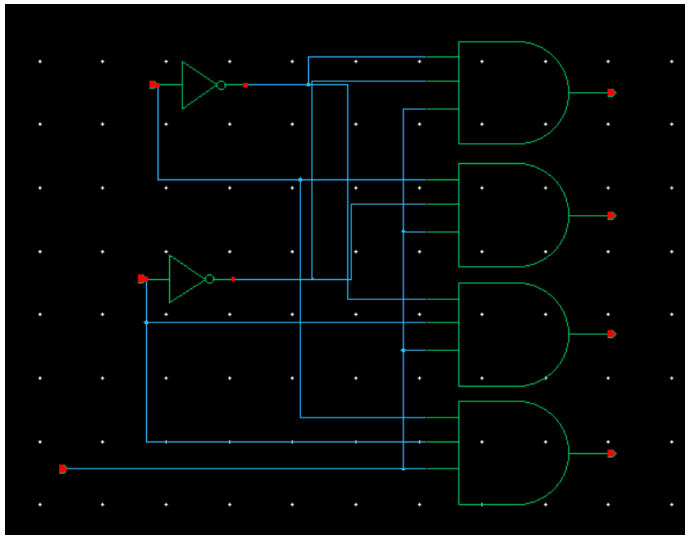


Figure 4: DECODER4 Schematic.

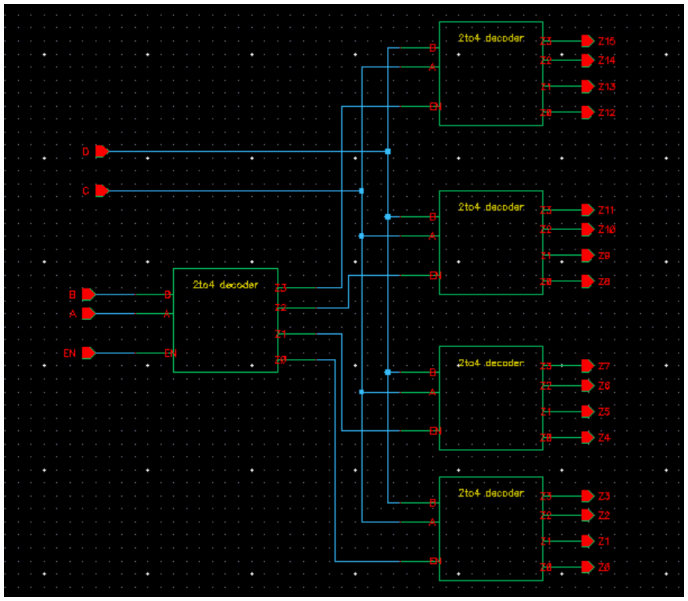


Figure 5: DECODER16 Schematic.

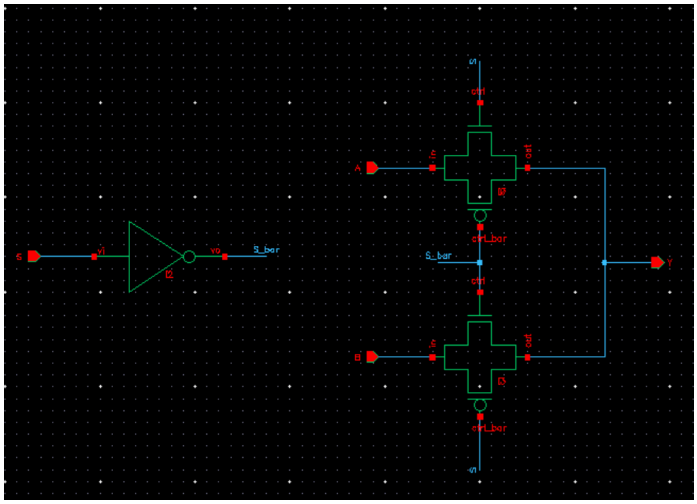


Figure 6: MUX2 Schematic.

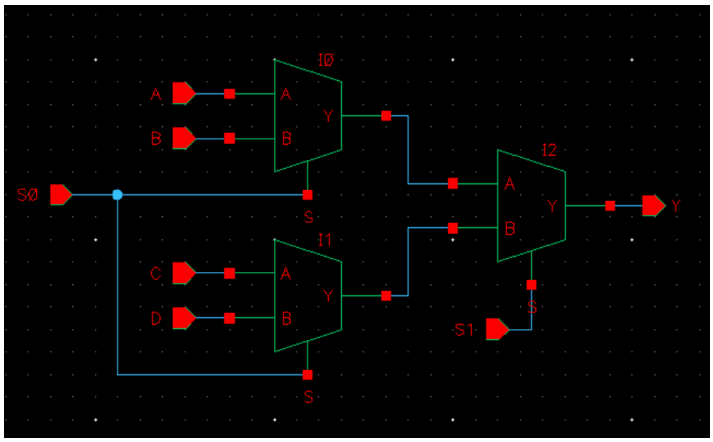


Figure 7: MUX4 Schematic.

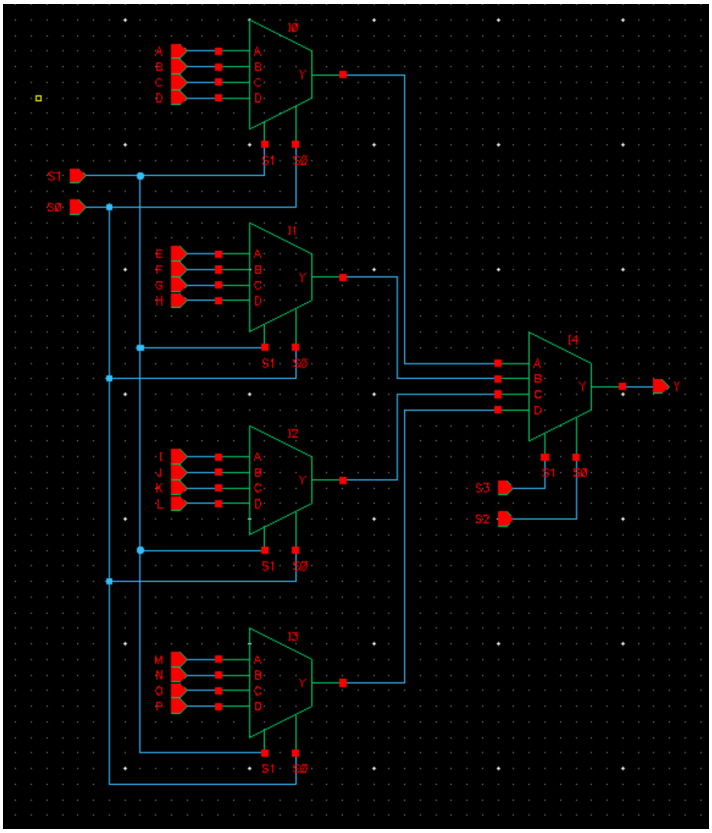


Figure 8: MUX16 Schematic.

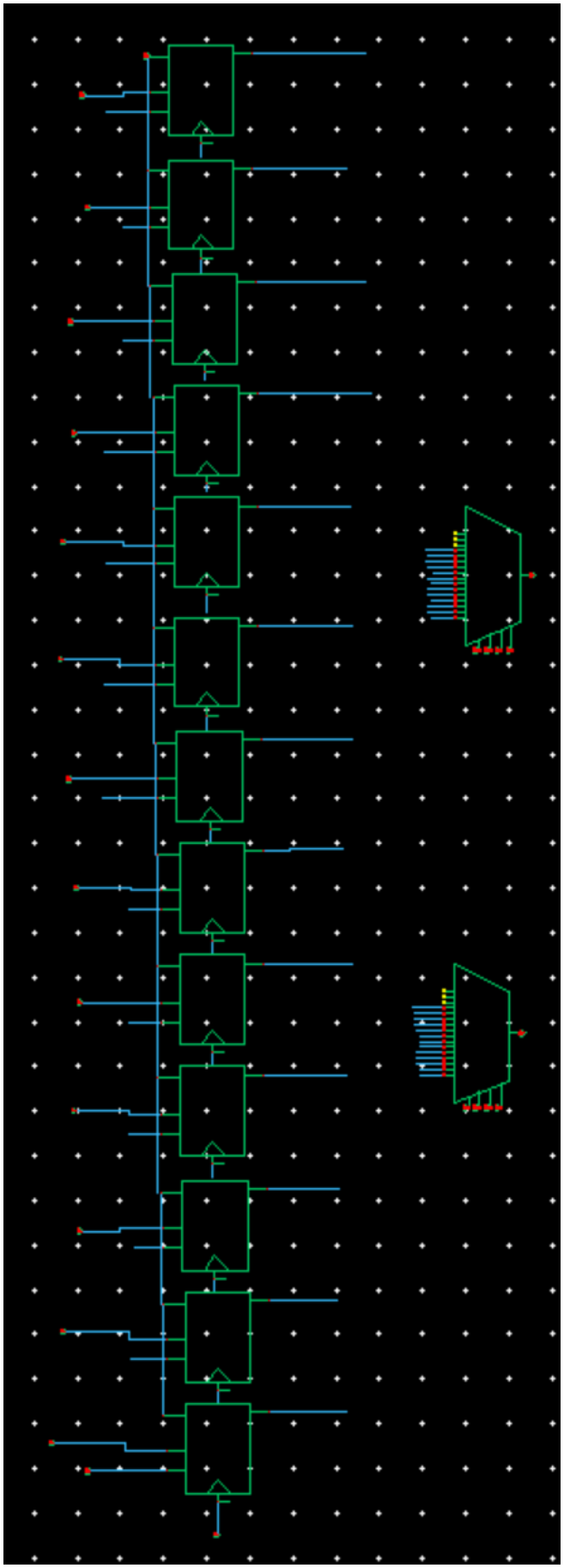


Figure 9: Bit Slice Schematic.

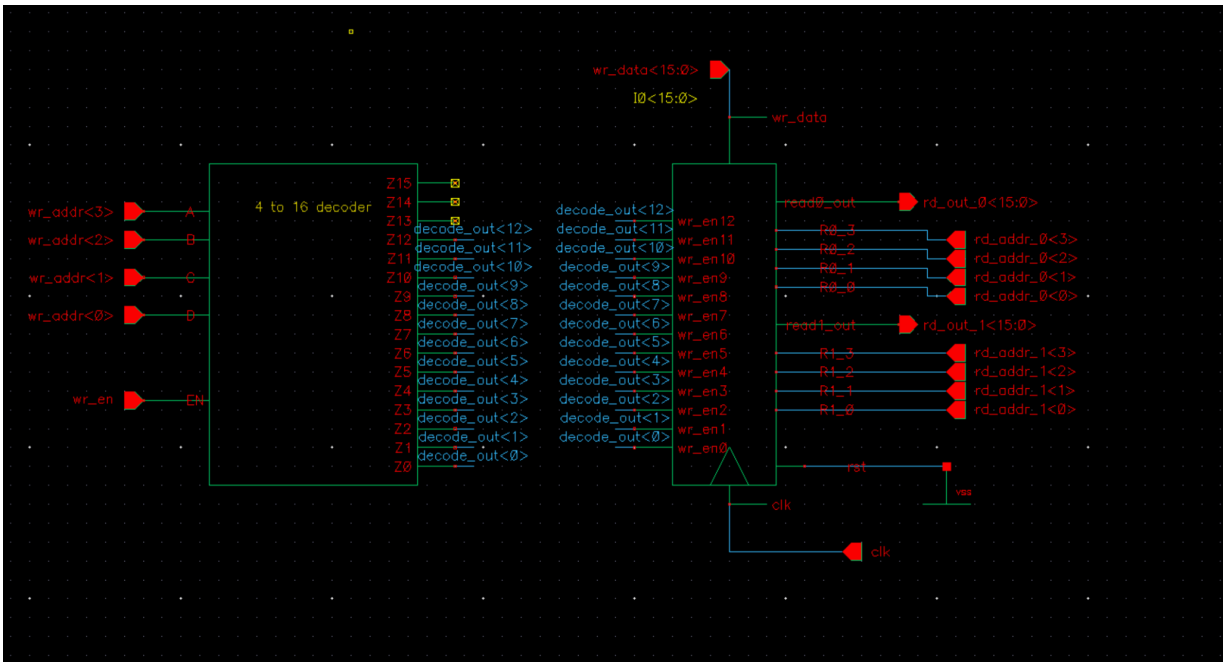


Figure 10: RF Schematic.

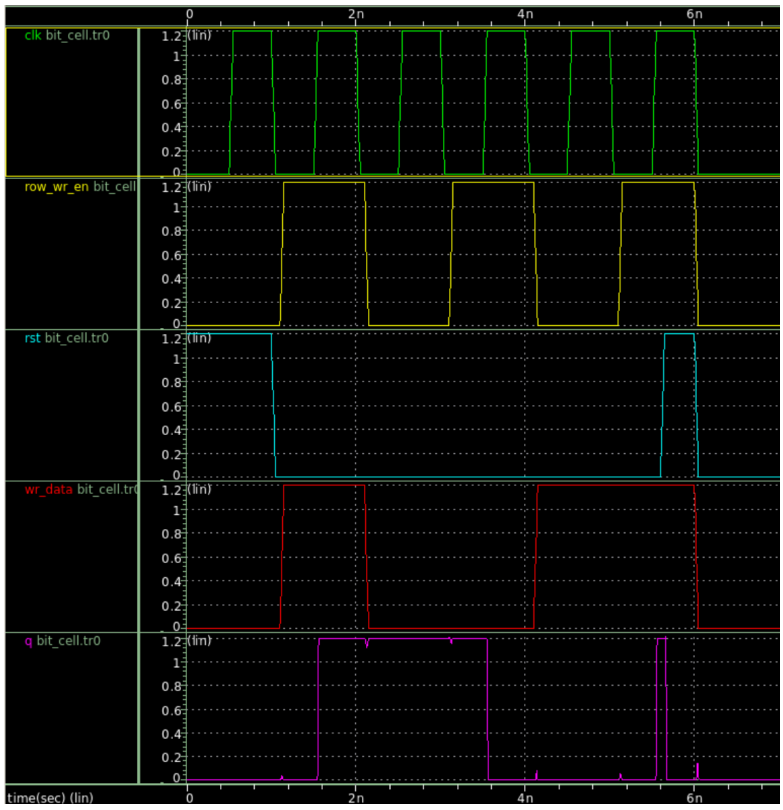


Figure 11: Bit Cell Testbench Waveform.

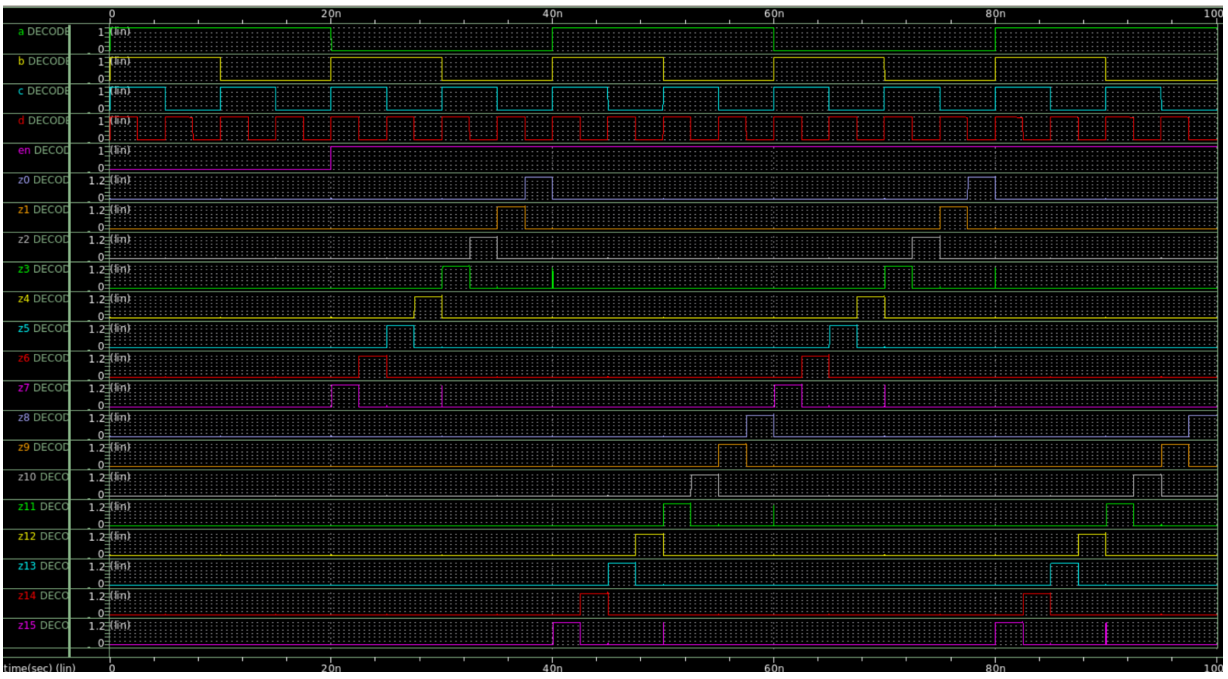


Figure 12: DECODER16 Testbench Waveform.

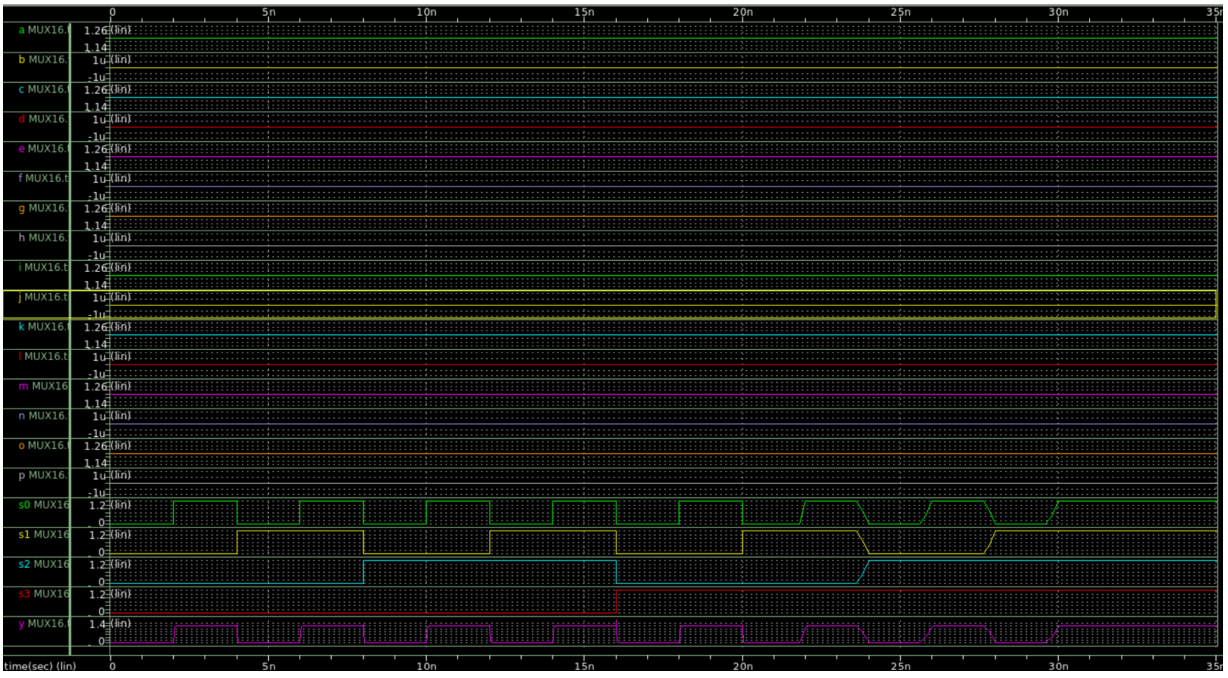


Figure 13: MUX16 Testbench Waveform.

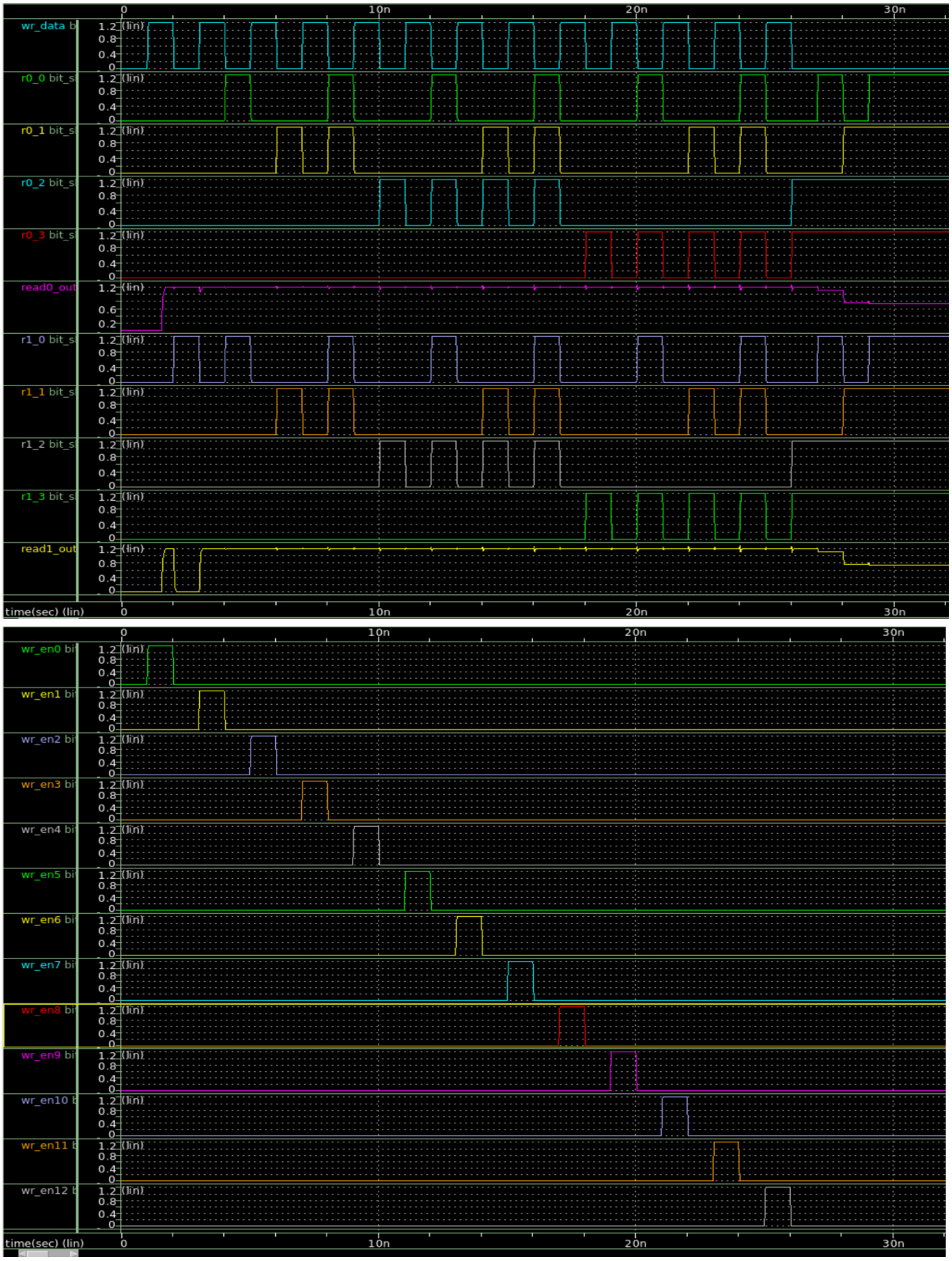


Figure 14: Bit Slice Testbench Waveform.